

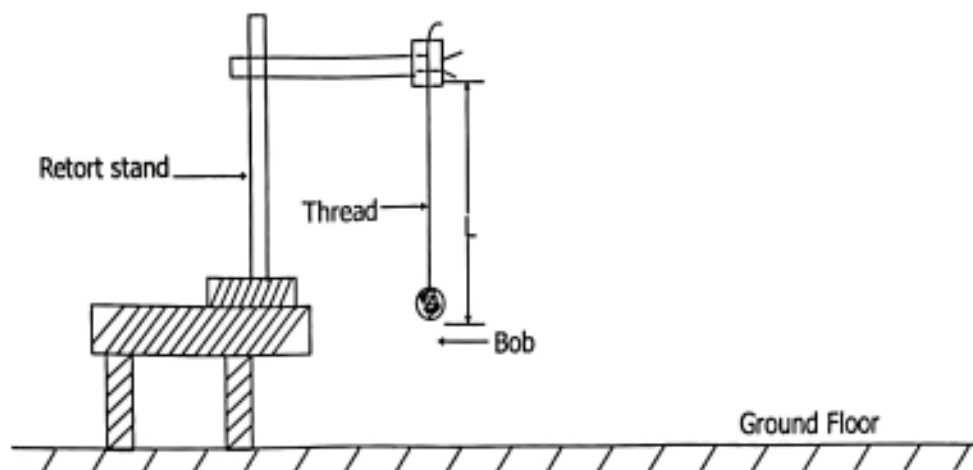
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DAR ES SALAAM REGION
FORM FOUR MOCK EXAMINATION - 2025
PHYSICS 2B (ACTUAL PRACTICAL)
GEPAM science hub
 WhatsApp 0748978398
 CODE: D31/zp
 Monday 28 July, 2025 A.M
 TIME: 2:30 Hours

INSTRUCTIONS

1. This paper consists of **two (2)** questions. Answer **all** two questions
2. Each question carries **twenty five (25)** marks.
3. All writings should be in blue or black ink, except for diagrams which must be in pencil.
4. Mathematical tables and non-programmable calculators may be used
5. Communication devices and any other unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet (s)

This paper consists of 3 printed pages

01. You are provided with pendulum bob, cotton thread, metre rule, stop watch and Retort stand proceed as follows:



- (i) Arrange the apparatus you provided as shown in the diagram above. Suspend a pendulum of length L equal to 20cm. Pull a side a pendulum bob at a small angle, release to make it swing to and fro, measure time ' t ' for fifteen (15) oscillations of the pendulum bob.
- (ii) Repeat above procedure for length ' L ' equal to 40cm, 60cm, 80cm and 100cm; each time record ' t ' for 15 oscillations.

(a) Tabulate your results for values of L (m), t (s), t^2 (s^2) and $\frac{t^2}{100}$ (s^2)

(b) Plot a graph of $\frac{t^2}{100}$ (s^2) against L (m)

(c) Determine slope, S the graph.

(d) Determine the value of X given that

$$\frac{t}{10} = \frac{2\pi n}{10} \sqrt{\frac{L}{X}}$$

Where n = is the number of oscillations.

(e) What does X represent?

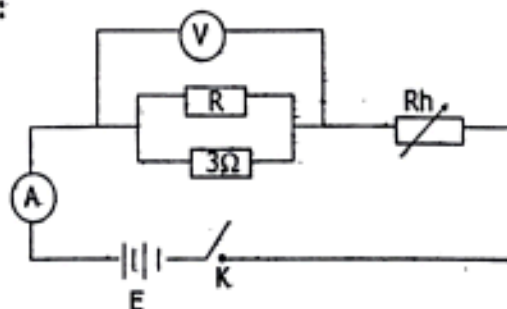
(f) Suggest the aim of this experiment.

(g) State the sources of errors and their precautions to be taken during the experiment.

(25 marks)

2. You are provided with two dry cells of e.m.f (E), a key ' K ', an ammeter (A), voltmeter (V), Rheostat (R_h), unknown resistor (R) and a 3Ω resistor

Proceed as follows:



- (a) Set up the circuit as shown in figure 2
- (b) Close the key and adjust the rheostat so that the ammeter reading is 0.1A. Record the voltmeter reading in volts.
- (c) Repeat the procedure in 2(b) for the values of ammeter reading of ,0.2A, 0.3A, 0.4A and 0.5A.

Questions

- (i) Tabulate your result including the values of $\frac{1}{I}$ and $\frac{1}{V} (V^{-1})$
- (ii) Plot a graph of $\frac{1}{I} (A^{-1})$ against $\frac{1}{V} (V^{-1})$
- (iii) Find the slope of the graph
- (iv) Determine the value of the unknown resistance. (R)
- (v) What is the effect of connecting resistor R as shown in figure 2? (25 marks)



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MARKING SCHEME

01. Q. TABLE OF RESULTS

L(m)	t(s)	t ² (s ²)	t ² /100 (s ²)
0.2	14	196	1.96
0.4	19	361	3.61
0.6	23	529	5.29
0.8	27	729	7.29
1.0	30	900	9.0

(001 marks @ 20 x 0.05 marks)

(b) A plotted graph of $\frac{t^2}{100}$ (s²) against L(m).

(c) Taking coordinates at (0.5, 4.5) and (0.95, 8.5)

$$\text{Slope} = \frac{\text{change in } \left(\frac{t^2}{100}\right)}{\text{change in } L} \quad (001 \frac{1}{2} \text{ marks})$$

$$\text{Slope} = \left(\frac{8.5 - 4.5}{0.95 - 0.5} \right) \frac{s^2}{m} \quad (001 \frac{1}{2} \text{ marks})$$

$$\text{Slope} = 8.89 \frac{s^2}{m} \quad (01 \text{ marks})$$

$$\text{Range } (8.7 - 9.2 \frac{s^2}{m})$$

$$(d) \frac{t}{10} = \frac{2\pi n}{10} \sqrt{\frac{L}{x}}$$

Square both sides, gives

$$\frac{t^2}{100} = \frac{4\pi^2 n^2 L}{100 x} \quad \text{--- (00\frac{1}{2} mark)}$$

$$\frac{t^2}{100} = \frac{4\pi^2 n^2 L}{100 x} \quad \text{but } n=15, \text{ substituting gives}$$

$$\frac{t^2}{100} = \frac{4\pi^2 \times 15^2 L}{100 x}$$

$$\frac{t^2}{100} = \left(\frac{900\pi^2}{100 x} \right) L \quad \text{--- (00\frac{1}{2} mark)}$$

Comparing with equation of straight line
 $y = mx + c$

$$\therefore \text{Slope} = \frac{900\pi^2}{100 x} \quad \text{--- (00\frac{1}{2} mark)}$$

$$x = \frac{900\pi^2}{100 \times \text{slope}}$$

$$x = \frac{900 \times (3.14)^2}{100 \times 8.89 \text{ s}^2/\text{m}} \quad \text{--- (00\frac{1}{2} mark)}$$

$$x = 9.98 \text{ m/s}^2 \quad \text{--- (02 marks)}$$

Range (9.65 $\rightarrow$ 10.2 m/s²).

1. (g) (e) The value of x represents the acceleration due to gravity, g . (01 marks)

(f) The aim of this experiment is to determine acceleration due to gravity, g . (01 marks)

(g) Sources of errors.

1. Parallax error. --- (00½ mark)

— Read a mark on the metre rule straight vertically from above (i.e. perpendicular to the scale of the pointer reading). --- (01 marks)

2. Air resistance (wind blowing). --- (00½ mark)

— Close the window or doors to block the air to reach the experiment. (01 marks)

02. (i) TABLE OF RESULTS

$I(A)$	$V(V)$	$\frac{1}{I}(A^{-1})$	$\frac{1}{V}(V^{-1})$
0.1	0.2	10.00	5.00
0.2	0.35	5.00	2.86
0.3	0.50	3.33	2.00
0.4	0.75	2.50	1.33
0.5	1.00	1.00	1.00

(10 marks)

(01 marks = 0.5 marks) (02 marks) (00.5 marks = 0.25 marks)

(ii) A plotted graph of $\frac{1}{I}(A^{-1})$ against $\frac{1}{V}(V^{-1})$ is on the graph paper.

(iii) Taking coordinates at (1.6, 2.4) and (2.4, 4.2)

$$\text{Slope} = \frac{\text{change in } \frac{1}{I}(A^{-1})}{\text{change in } \frac{1}{V}(V^{-1})} \dots \dots (00.5 \text{ mark})$$

$$\text{Slope} = \left(\frac{4.2 - 2.4}{2.4 - 1.6} \right) \frac{A^{-1}}{V^{-1}} \dots \dots (00.5 \text{ mark})$$

$$\text{Slope} = 2.25 \frac{A^{-1}}{V^{-1}}$$

$$\therefore \text{Slope} = 2.25 \Omega \dots \dots (01 \text{ marks})$$

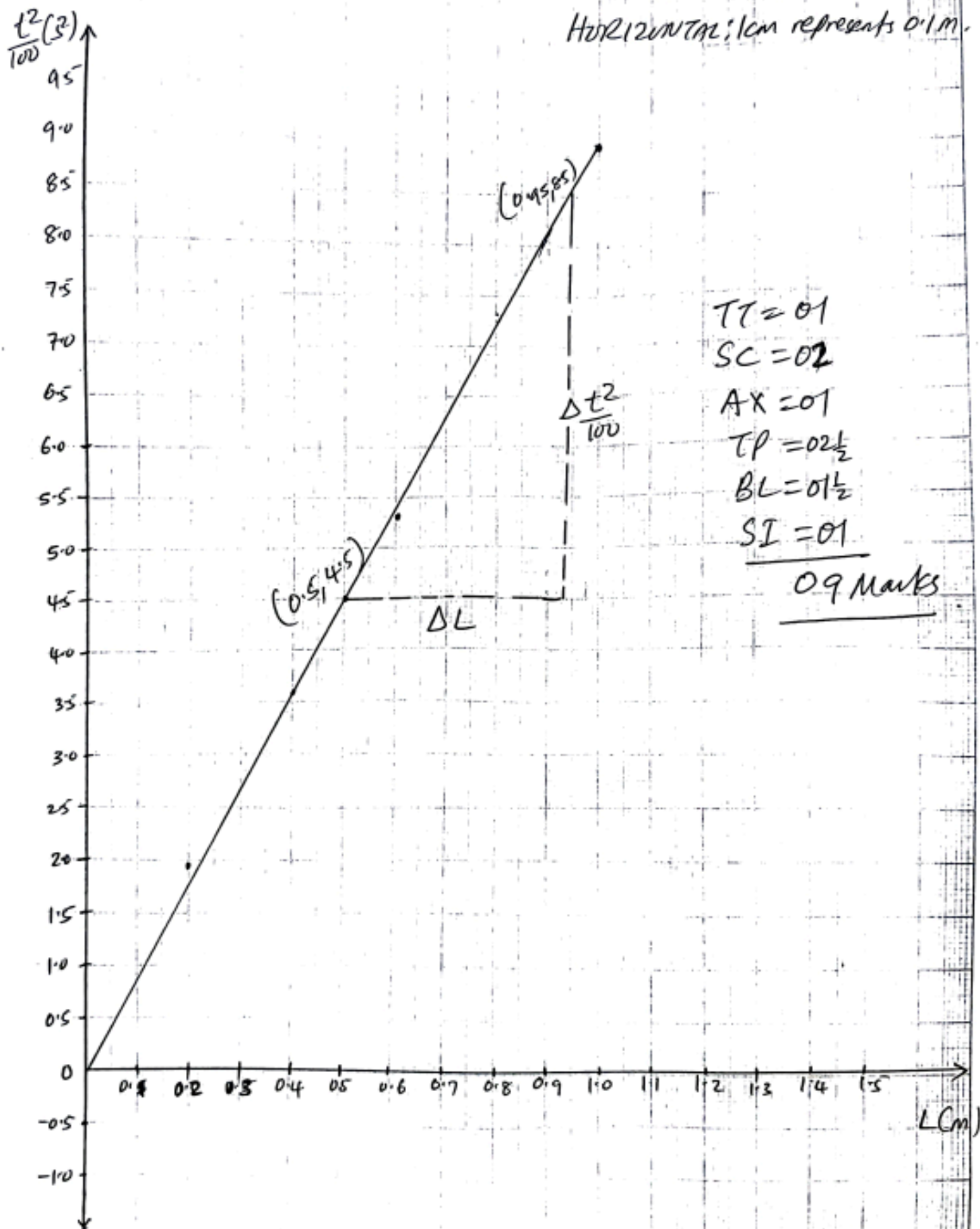
(iv) The unknown resistance equals to the slope of the graph, $R = 2.25 \Omega$. (01 marks)

(v) Connecting resistor R in parallel with 3Ω resistor is to reduce the value of resistance in the circuit. (02 marks)

(b)

A GRAPH OF $\frac{t^2}{100} (s^2)$ AGAINST $L (m)$

SCALE:

VERTICAL: 1cm represents $0.5s^2$ HORIZONTAL: 1cm represents $0.1m$ 

$$TT = 01$$

$$SC = 02$$

$$AX = 01$$

$$TP = 02\frac{1}{2}$$

$$BL = 01\frac{1}{2}$$

$$SI = 01$$

09 Marks

Qn. 2 (ii)

A GRAPH OF $\frac{1}{I} (A')$ AGAINST $\frac{1}{V} (V')$.

SCALE:

VERTICAL: 1cm represents 0.6 A'

HORIZONTAL: 1cm represents 0.4 V'

